Git Laboratory Practice Set

Questions 3 & 4

QUESTION 3 — WebPortal Project

Problem 1: Initialize & First Commit

1. Create a folder named WebPortal.
2. Configure Git with your username and email.
3. Initialize the folder as a Git repository.
4. Create index.html containing: Welcome to Web Portal.
5. Stage and commit with message: Initial WebPortal commit.
6. Show the repository status and commit history.

Problem 2: Multiple Files & History

1. Create style.css and script.js.
2. Add any content to both files.
3. Stage only style.css.
4. Check the repository status and observe the difference.
5. Commit style.css with a meaningful message.
6. Stage and commit script.js separately.
7. View the commit history and explain each commit.

Problem 3: Reset to Previous Version

1. Edit index.html by adding: Homepage Updated.
2. Stage and commit the change.
3. Copy the ID of the first commit.
4. Reset the repository to the first commit.
5. Verify that the updated line has been removed.

Problem 4: Frontend Branch

1. Create a branch named frontend.
2. Switch to the branch.
3. Create navbar.html containing: Navigation Bar.
4. Stage and commit the file.
5. Verify that the commit exists only in frontend.

Problem 5: Backend Branch & Merge

1. Create a branch named backend.
2. Add api.txt containing: Backend API Ready.
3. Commit the file.
4. Merge frontend into master.
5. Merge backend into master.
6. Verify that both files exist.

Problem 6: Parallel Changes & Conflict

1. Modify index.html on master and commit.
2. Modify the same line differently on backend and commit.
3. Merge backend into master.
4. Observe and explain the merge conflict.

QUESTION 4 — LibrarySystem Project

Problem 1: Setup Repository

1. Create a folder named LibrarySystem.
2. Initialize it as a Git repository.
3. Configure your Git username.
4. Create catalog.txt containing: Library Catalog.
5. Stage and commit the file.

Problem 2: Tracking Multiple Files

1. Create members.txt and books.txt.
2. Add any content to both files.
3. Stage only members.txt.
4. Check status and observe the difference.
5. Commit members.txt.
6. Stage and commit books.txt separately.
7. Explain the commit history.

Problem 3: Restore Previous Version

1. Edit catalog.txt by adding: Digital Catalog Enabled.
2. Stage and commit the change.
3. Return to the first commit.

4. Verify that the added line is removed.

Problem 4: Inventory Branch

1. Create branch inventory.
2. Switch to it.
3. Create stock.txt containing: Stock Management Started.
4. Commit the file.
5. Verify the branch history.

Problem 5: Issue System Merge

1. Create branch issue-system.
2. Add issue.txt containing: Book Issue Module.
3. Commit the file.
4. Merge inventory and issue-system into master.
5. Verify both files exist.

Problem 6: Parallel Development

1. Edit catalog.txt on master and commit.
2. Edit the same file differently on inventory and commit.
3. Merge inventory into master.
4. Observe and explain Git's merge behavior.